

Vector Spaces

Def 1.1 A vector space V over a field F is set of vectors that is closed under finite vector addition and scalar multiplication, and also satisfies the following eight axioms

For any $\vec{u}, \vec{v}, \vec{w} \in V$ and $a, b \in F$

1.1.i. $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$

1.1.ii. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

1.1.iii. There exists a zero vector $\vec{0} \in V$ s.t.
 $\vec{0} + \vec{v} = \vec{v}$

1.1.iv. There exist additive inverse $-\vec{v} \in V$
s.t. $\vec{v} + (-\vec{v}) = \vec{0}$

1.1.v. $a(b\vec{v}) = (ab)\vec{v}$

1.1.vi. There exists mult. identity $1 \in F$
s.t. $1 \cdot \vec{v} = \vec{v}$

1.1.vii. $(a+b)\vec{v} = a\vec{v} + b\vec{v}$

1.1.viii. $a(\vec{u} + \vec{v}) = a\vec{u} + a\vec{v}$

algebraic

Examples of fields: $\mathbb{R}, \mathbb{C}, \mathbb{Q}(\sqrt{2})$, $\mathbb{Z}/p\mathbb{Z}$
extension
p-prime

Exercise

1) Determine whether or not the set

$$S = \{ (x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 \leq 4 \}$$

is a vector space.

No! As $\vec{x} = (2, 0, 0) \in S$ and

$2 \cdot \vec{x} \notin S$, S is not closed under scalar mult. Also as

$(2, 0, 0) + (0, 2, 0) \notin S$, S is not closed under vector addition.

* Here we have that $S \subseteq \mathbb{R}^3$ but
↓
subset
not a subspace

Examples of Vectors Spaces

Finite dimension

i. $\mathbb{R}^n = \{ (x_1, \dots, x_n) : x_i \in \mathbb{R} \}$

ii. $\mathbb{C}^n = \{ (z_1, \dots, z_n) : z_i \in \mathbb{C} \}$

- iii. $P_n(F) = \{ a_0 + a_1x + \dots + a_nx^n : a_i \in F \}$
- iv. $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ - set of linear maps from $\mathbb{R}^m$ to $\mathbb{R}^n$
- v. $F^{m \times n}$ - set of $m \times n$ matrices over F

Infinite dimension vector spaces

- i. Lebesgue space: ℓ^p, L^p
- ii. Bergman and Fock spaces
- iii. Hilbert Space of quantum states

Euclidean Space $\mathbb{R}^n$

$\mathbb{R}^n$ is the vector space of n -tuples of $\mathbb{R}$

For $\vec{x}, \vec{y} \in \mathbb{R}^n$ where $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$ we have that

$$\vec{x} + \vec{y} = (x_1 + y_1, \dots, x_n + y_n)$$

$$\lambda \vec{x} = (\lambda x_1, \dots, \lambda x_n)$$

$\mathbb{R}^n$ has the following nice properties:

- i. It is complete

ii. It is normed

iii. It has an inner product

Property i. comes from the completeness of $\mathbb{R}$ (no holes) Properties i. and ii. mean that $\mathbb{R}^n$ is a Banach space and with property iii. we have that $\mathbb{R}^n$ is a Hilbert space.

Inner Product on $\mathbb{R}^n$ (dot product)

For $\vec{x}, \vec{y} \in \mathbb{R}^n$ we define their inner product as

$$\langle \vec{x}, \vec{y} \rangle = \sum_{i=1}^n x_i y_i$$

Properties of $\langle \cdot, \cdot \rangle$ for $F = \mathbb{C}$ or $\mathbb{R}$

1.2.i $\langle \vec{x}, \vec{y} \rangle = \overline{\langle \vec{y}, \vec{x} \rangle}$

in $\mathbb{R}^n$: $\langle \vec{x}, \vec{y} \rangle = \langle \vec{y}, \vec{x} \rangle$

1.2.ii $\lambda \langle \vec{x}, \vec{y} \rangle = \langle \lambda \vec{x}, \vec{y} \rangle = \langle \vec{x}, \bar{\lambda} \vec{y} \rangle$

in $\mathbb{R}^n$: $\langle \lambda \vec{x}, \vec{y} \rangle = \langle \vec{x}, \lambda \vec{y} \rangle$

$$1.2.iii \quad \langle \lambda_1 \vec{x}_1 + \lambda_2 \vec{x}_2, \vec{y} \rangle = \lambda_1 \langle \vec{x}_1, \vec{y} \rangle + \lambda_2 \langle \vec{x}_2, \vec{y} \rangle$$

in $\mathbb{R}^n$: $\langle \cdot, \cdot \rangle$ is bilinear

Orthogonality: $\vec{x}, \vec{y} \in \mathbb{R}^n$ are orthogonal
iff $\langle \vec{x}, \vec{y} \rangle = 0$ (true for any inner product space)

Given an inner product $\langle \cdot, \cdot \rangle$ we define the norm as $\|\vec{x}\| = \sqrt{\langle \vec{x}, \vec{x} \rangle}$

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow F$$

$$\|\cdot\| : V \rightarrow [0, \infty)$$

Def 1.3 Given $\vec{x} \in \mathbb{R}^n$ we define the Euclidean norm of $\vec{x}$ as $\|\vec{x}\| = \sqrt{\langle \vec{x}, \vec{x} \rangle} = \left(\sum_i x_i^2 \right)^{1/2}$

* This known as the l^2 norm

Properties of $\langle \cdot, \cdot \rangle$ and $\|\cdot\|$ on $\mathbb{R}^n$

$$1.3.i. \quad \|\vec{x}\| = 0 \text{ iff } \vec{x} = 0$$

$$1.3.ii \quad \|\lambda \vec{x}\| = |\lambda| \|\vec{x}\|$$

$$1.3.iii \quad |\langle \vec{x}, \vec{y} \rangle| \leq \|\vec{x}\| \|\vec{y}\| \text{ (Cauchy-Schwarz)}$$

$$\text{in } \mathbb{R}^2: \quad \langle \vec{x}, \vec{y} \rangle = \|\vec{x}\| \|\vec{y}\| \cos \theta$$

$$1.3.iv. \quad \|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\| \text{ (triangle inequality)}$$

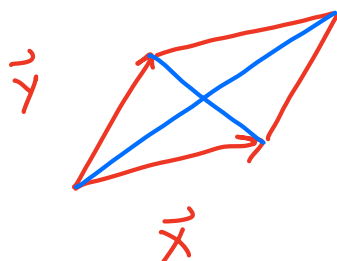
$$1.3.v. \quad \langle \vec{x}, \vec{x} \rangle = \|\vec{x}\|^2$$

$$1.3.vi. \quad \langle \vec{x} - \vec{y}, \vec{x} + \vec{y} \rangle = \|\vec{x}\|^2 - \|\vec{y}\|^2$$

$$\text{in } \mathbb{C}^n: \quad \langle \vec{x} - \vec{y}, \vec{x} + \vec{y} \rangle = \|\vec{x}\|^2 - \|\vec{y}\|^2 + 2 \operatorname{Im} \langle \vec{x}, \vec{y} \rangle$$

$$1.3.vii. \quad \|\vec{x} + \vec{y}\|^2 + \|\vec{x} - \vec{y}\|^2 = 2(\|\vec{x}\|^2 + \|\vec{y}\|^2)$$

Parallelogram law



Exercises:

2) Show that if $\|\vec{x}\| = \|\vec{y}\| = 1$ and $\langle \vec{x}, \vec{y} \rangle = 1$ then $\vec{x} = \vec{y}$.

For $\mathbb{R}^2$ use law of cosines. Now for $\vec{x}, \vec{y} \in \mathbb{R}^n$ where $n > 2$ we

will show that $\vec{x} - \vec{y} = \vec{0}$. Since

$$\begin{aligned}\|\vec{x} - \vec{y}\|^2 &= \langle \vec{x} - \vec{y}, \vec{x} - \vec{y} \rangle^* \\ &= \|\vec{x}\|^2 - 2\langle \vec{x}, \vec{y} \rangle + \|\vec{y}\|^2 \\ &= 1 - 2 + 1 = 0\end{aligned}$$

$$\begin{aligned}^* \langle \vec{x} - \vec{y}, \vec{x} - \vec{y} \rangle &= \langle \vec{x}, \vec{x} - \vec{y} \rangle - \langle \vec{y}, \vec{x} - \vec{y} \rangle \\ &= \langle \vec{x}, \vec{x} \rangle - \langle \vec{x}, \vec{y} \rangle - \langle \vec{y}, \vec{x} \rangle \\ &\quad + \langle \vec{y}, \vec{y} \rangle\end{aligned}$$

we have that $\vec{x} - \vec{y} = \vec{0}$ by 1.3.i
and thus $\vec{x} = \vec{y}$.

3) Prove Cauchy - Schwarz for $\mathbb{R}^3$ using
the inequality, $a^2 + b^2 \geq 2ab$.

$$(a-b)^2 \geq 0$$

We will show that $\langle \vec{x}, \vec{y} \rangle^2 \leq \|\vec{x}\|^2 \|\vec{y}\|^2$.

Using the above identity we have that

$$\begin{aligned}\langle \vec{x}, \vec{y} \rangle^2 &= (x_1 y_1 + x_2 y_2 + x_3 y_3)^2 \\ &= x_1^2 y_1^2 + x_2^2 y_2^2 + x_3^2 y_3^2 \\ &\quad + 2(x_1 y_2)(x_2 y_1) + 2(x_1 y_3)(x_3 y_1) \\ &\quad + 2(x_2 y_3)(x_3 y_2)\end{aligned}$$

$$\begin{aligned}
&\leq \underbrace{x_1^2 y_1^2} + \underbrace{x_2^2 y_2^2} + \underbrace{x_3^2 y_3^2} \\
&\quad + \underbrace{x_1^2 y_2^2} + \underbrace{x_2^2 y_1^2} + \underbrace{x_1^2 y_3^2} + \underbrace{x_3^2 y_1^2} \\
&\quad + \underbrace{x_2^2 y_3^2} + \underbrace{x_3^2 y_2^2} \\
&= \underbrace{x_1^2} \|\vec{y}\|^2 + \underbrace{x_2^2} \|\vec{y}\|^2 + \underbrace{x_3^2} \|\vec{y}\|^2 \\
&= \|\vec{x}\|^2 \|\vec{y}\|^2
\end{aligned}$$

as desired